

CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

# Chapter 1: Orienting Yourself: The Use of Coordinates

Topic-wise short notes with solved examples

Two numbers are enough to fix the position of any point in a plane. This chapter sets up the coordinate plane and uses it to find the distance between two points.

## 1 The Cartesian plane

- A **coordinate system** uses numbers to describe the exact position of a point.
- Two number lines are drawn at right angles. The horizontal line is the  **$x$ -axis** and the vertical line is the  **$y$ -axis**. Together they are the **coordinate axes**.
- The axes meet at the **origin**  $O$ , whose coordinates are  $(0, 0)$ .
- Equal units are marked on both axes. Distances to the **right** of  $O$  and **upwards** from  $O$  are positive. Distances to the **left** and **downwards** are negative.
- The plane containing the axes is called the **Cartesian plane**, the **coordinate plane** or the  **$xy$ -plane**.

### REMEMBER

The system is named after René Descartes (1637). The negative half of each axis depends on Brahmagupta's work on zero and negative numbers (628 CE).

## 2 Coordinates of a point

- Every point  $P$  in the plane is written as  $P(x, y)$ .
- The  **$x$ -coordinate** is the perpendicular distance of  $P$  from the  $y$ -axis, measured along the  $x$ -axis.
- The  **$y$ -coordinate** is the perpendicular distance of  $P$  from the  $x$ -axis, measured along the  $y$ -axis.
- The order matters. If  $x \neq y$ , then  $(x, y) \neq (y, x)$ . The two are the same point only when  $x = y$ .
- A point on the  $x$ -axis has the form  $(x, 0)$ . A point on the  $y$ -axis has the form  $(0, y)$ .

**Example 2.1** Write the coordinates of the point that is (a) on the  $x$ -axis, 4.5 units to the right of  $O$  (b) on the  $y$ -axis, 4.5 units below  $O$ .

### SOLUTION

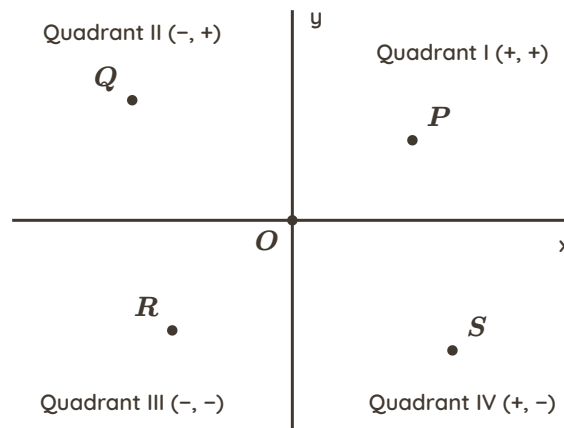
(a) It is on the  $x$ -axis, so  $y = 0$ . To the right means  $x$  is positive.

(b) It is on the  $y$ -axis, so  $x = 0$ . Below means  $y$  is negative.

**(a)  $(4.5, 0)$  (b)  $(0, -4.5)$**

### 3 The four quadrants

- The axes divide the plane into four parts called **quadrants**, numbered anticlockwise from the top right.



$P(3, 2)$ ,  $Q(-4, 3)$ ,  $R(-3, -3)$  and  $S(4, -3.5)$  lie in the four quadrants

| Quadrant | Sign of $x$ | Sign of $y$ | Example    |
|----------|-------------|-------------|------------|
| I        | +           | +           | $(3, 2)$   |
| II       | -           | +           | $(-5, 3)$  |
| III      | -           | -           | $(-3, -3)$ |
| IV       | +           | -           | $(3, -5)$  |

- A point on an axis does not belong to any quadrant.

**Example 3.1** In which quadrant, or on which axis, does each point lie:  $(-5, 3)$ ,  $(3, -5)$ ,  $(0, 4)$ ,  $(-2.9, 0)$ ?

**SOLUTION**

$(-5, 3)$ :  $x$  negative,  $y$  positive.  $(3, -5)$ :  $x$  positive,  $y$  negative.

$(0, 4)$  has  $x = 0$  and  $(-2.9, 0)$  has  $y = 0$ .

**Quadrant II; Quadrant IV; on the  $y$ -axis; on the  $x$ -axis**

### 4 Distance along a line parallel to an axis

- Two points with the same  $y$ -coordinate lie on a line parallel to the  $x$ -axis. The distance between  $(x_1, y)$  and  $(x_2, y)$  is  $|x_2 - x_1|$ .
- Two points with the same  $x$ -coordinate lie on a line parallel to the  $y$ -axis. The distance between  $(x, y_1)$  and  $(x, y_2)$  is  $|y_2 - y_1|$ .
- The bars mean the **absolute value**: take the difference and ignore its sign.

**Example 4.1** A door has its two ends at  $(8.5, 0)$  and  $(11.5, 0)$ . How wide is it? Another door has ends at  $(0, 1.5)$  and  $(0, 4)$ . Which door is wider?

SOLUTION

First door:  $|11.5 - 8.5| = 3$  units. Second door:  $|4 - 1.5| = 2.5$  units.

**The first door is 3 units wide and is the wider one.**

**Example 4.2** Three corners of a rectangular table are at  $(8, 9)$ ,  $(11, 9)$  and  $(11, 7)$ . Find the fourth corner and the length and width of the table.

SOLUTION

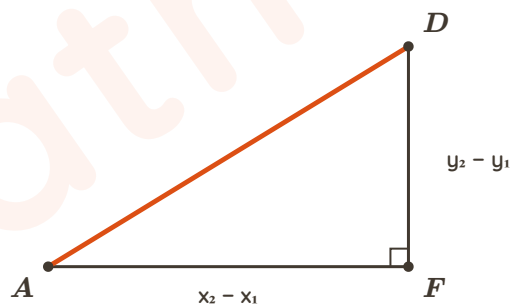
The fourth corner has the  $x$ -coordinate of the first point and the  $y$ -coordinate of the third.

Length =  $|11 - 8| = 3$  units and width =  $|9 - 7| = 2$  units.

**Fourth corner  $(8, 7)$ ; the table is 3 units by 2 units.**

## 5 The distance formula

- For a slanting segment, draw a right-angled triangle with sides parallel to the axes and use the Baudhāyana-Pythagoras theorem.



AD is the hypotenuse of a right-angled triangle

- The distance between  $A(x_1, y_1)$  and  $D(x_2, y_2)$  is

$$AD = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- It makes no difference whether  $x_2 - x_1$  and  $y_2 - y_1$  are positive or negative, because they are squared.
- The distance of  $(x, y)$  from the origin is  $\sqrt{x^2 + y^2}$ .

**Example 5.1** Find the lengths of the sides of the triangle with vertices  $A(3, 4)$ ,  $D(7, 1)$  and  $M(9, 6)$ .

SOLUTION

$$AD = \sqrt{(7-3)^2 + (1-4)^2} = \sqrt{16+9} = 5 \text{ units}$$

$$DM = \sqrt{(9-7)^2 + (6-1)^2} = \sqrt{4+25} = \sqrt{29} \text{ units}$$

$$MA = \sqrt{(3-9)^2 + (4-6)^2} = \sqrt{36+4} = \sqrt{40} \text{ units}$$

$$AD = 5, DM = \sqrt{29} \text{ and } MA = \sqrt{40} \text{ units}$$

**Example 5.2** Show that  $A(2, 1)$ ,  $B(-1, 2)$ ,  $C(-2, -1)$  and  $D(1, -2)$  are the vertices of a square, and find its area.

SOLUTION

$AB^2 = 9 + 1 = 10$ ,  $BC^2 = 1 + 9 = 10$ ,  $CD^2 = 9 + 1 = 10$ ,  $DA^2 = 1 + 9 = 10$ , so all four sides are equal.

Diagonals:  $AC^2 = 16 + 4 = 20$  and  $BD^2 = 4 + 16 = 20$ , so the diagonals are equal.

Four equal sides and equal diagonals make a square. Area =  $(\sqrt{10})^2$ .

**$ABCD$  is a square of area 10 square units.**

## 6 Reflection in the axes

- Reflecting a point in the  **$y$ -axis** changes the sign of its  $x$ -coordinate:  $(x, y) \rightarrow (-x, y)$ .
- Reflecting a point in the  **$x$ -axis** changes the sign of its  $y$ -coordinate:  $(x, y) \rightarrow (x, -y)$ .
- A reflection does not change lengths. A triangle and its image have equal sides.

**Example 6.1** Triangle  $ADM$  with  $A(3, 4)$ ,  $D(7, 1)$ ,  $M(9, 6)$  is reflected in the  $y$ -axis. Write the images and check that  $A'D' = AD$ .

SOLUTION

$$A'(-3, 4), D'(-7, 1) \text{ and } M'(-9, 6)$$

$$A'D' = \sqrt{(-7+3)^2 + (1-4)^2} = \sqrt{16+9} = 5 \text{ units, the same as } AD.$$

**The reflection keeps the lengths of the sides.**

## 7 Using distances: collinear points, mid-points and circles

- **Collinear points:** three points lie on one line if the largest of the three distances between them equals the sum of the other two.
- **Mid-point:** the mid-point  $M$  of the segment joining  $S(x_1, y_1)$  and  $T(x_2, y_2)$  is

$$M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Circle:** a point lies on a circle with centre  $C$  and radius  $r$  if its distance from  $C$  equals  $r$ . It lies inside if the distance is less than  $r$  and outside if it is greater.

**Example 7.1** Are the points  $M(-3, -4)$ ,  $A(0, 0)$  and  $G(6, 8)$  on one straight line?

SOLUTION

$$MA = \sqrt{9 + 16} = 5, AG = \sqrt{36 + 64} = 10, MG = \sqrt{81 + 144} = 15$$

$$MA + AG = 5 + 10 = 15 = MG$$

**Yes, the three points are collinear.**

**Example 7.2**  $M(-7, 1)$  is the mid-point of  $A(3, -4)$  and  $B(x, y)$ . Find  $B$ .

SOLUTION

$$\frac{3 + x}{2} = -7 \text{ gives } x = -17, \text{ and } \frac{-4 + y}{2} = 1 \text{ gives } y = 6.$$

$$B = (-17, 6)$$

**Example 7.3** Show that  $A(1, -8)$ ,  $B(-4, 7)$  and  $C(-7, -4)$  lie on a circle with centre at the origin. Does  $D(-5, 6)$  lie inside, on or outside this circle?

SOLUTION

$$OA = \sqrt{1 + 64} = \sqrt{65}, OB = \sqrt{16 + 49} = \sqrt{65}, OC = \sqrt{49 + 16} = \sqrt{65}$$

All three are at the same distance from  $O$ , so they lie on a circle of radius  $\sqrt{65}$ .

$$OD = \sqrt{25 + 36} = \sqrt{61}, \text{ which is less than } \sqrt{65}.$$

**The radius is  $\sqrt{65}$  units, and  $D$  lies inside the circle.**

## 8 Common mistakes

- Writing the coordinates in the wrong order. The  $x$ -coordinate always comes first.
- Thinking a point on an axis is in a quadrant. It is on the boundary and belongs to none.
- Sign errors with negative coordinates:  $-3 - (-7) = 4$ , not  $-10$ .
- Forgetting to square before adding in the distance formula, or forgetting the square root at the end.
- Saying three points are collinear because two of the distances are equal. Check that two distances **add up** to the third.